

Studying the effect of non-spherical micro-particles on Hoek–Brown strength parameter m_i using numerical true triaxial compressive tests

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SUMMARY

The strength parameter m_i in the Hoek–Brown strength criterion is empirical and was developed by trial and error. To better understand the fundamental relationship between m_i and the physical characteristics of intact rock, this paper presents a systematic study of m_i by representing intact rock as a densely packed cemented particle material and simulating its mechanical behavior using particle flow modeling. Specifically, the three-dimensional particle flow code (PFC3D) was used to conduct numerical true triaxial compression tests on intact rock and to investigate the effect of non-spherical micro-particle parameters on m_i . To generate numerical intact rock specimens containing non-spherical micro-particles, a new genesis process was proposed, and a specific loop algorithm was used based on the efficiency of the process and the acceptability of generated specimens. Four main parameters—number, aspect ratio, size, and shape—of non-spherical micro-particles were studied, and the results indicated that they all have great effect on m_i . The strength parameter m_i increases when the number, aspect ratio, or size is larger or the shape becomes more irregular, mainly as a result of the higher level of interlocking between particles. This confirms the observations from engineering experience and laboratory experiments. To simulate the right strength parameter m_i , it is important to use appropriate non-spherical micro-particles by controlling these four parameters. This is further demonstrated by the simulation of two widely studied rocks, Lac du Bonnet granite and Carrara marble. Copyright © 2014 John Wiley & Sons, Ltd.

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1. INTRODUCTION

The Hoek–Brown (H–B) strength criterion was originally developed for intact rock and then extended to rock masses. The process used by Hoek and Brown in deriving their strength criterion for intact rock was one of pure trial and error. The justification for choosing this particular criterion over the numerous alternatives lies in the adequacy of its predictions of the observed rock fracture behavior, and the convenience of its application to a range of typical engineering problems. For intact rock, the H–B strength criterion may be expressed in the following form [1]

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_i \frac{\sigma_3}{\sigma_c} + 1 \right)^{0.5} \quad (1)$$

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where σ_c is the unconfined compressive strength of the intact rock, σ_1 and σ_3 are respectively the major and minor effective principal stresses, and m_i is a material constant for the intact rock. It is noted that the criterion is mainly determined by the strength parameter m_i .

To take account of the effect of the intermediate principal stress, Zhang and Zhu [2] proposed a three-dimensional (3D) version of the H-B strength criterion for intact rock and rock mass by combining the general Mogi [3] criterion and the H-B strength criterion. For intact rock, the 3D version of the H-B strength criterion is expressed as

$$\frac{9}{2\sigma_c}\tau_{\text{oct}}^2 + \frac{3}{2\sqrt{2}}m_i\tau_{\text{oct}} - m_i\sigma_{m,2} = \sigma_c \quad (2)$$

where τ_{oct} is the octahedral shear stress as defined by

$$\tau_{\text{oct}} = \frac{1}{3}\sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2} \quad (3)$$

and $\sigma_{m,2}$ is the mean stress defined by

$$\sigma_{m,2} = \frac{\sigma_1 + \sigma_3}{2} \quad (4)$$

in which σ_1 , σ_2 , and σ_3 are, respectively, the major, intermediate, and minor effective principal stresses. This 3D strength criterion inherits the advantages of the original H-B strength criterion in that it uses the same input parameters.

Hoek and Brown [4] stated that the strength parameter m_i is dimensionless and its value is affected by inter-particle friction and the degree of particle interlocking. In a latter version of the H-B strength criterion, Marinos and Hoek [5] further stated that the value of m_i depends on the granularity and interlocking of crystal structure, higher m_i values associated with more tightly interlocked and more frictional characteristics. It is noted that these conclusions are simply based on experience gained from engineering and laboratory tests. The m_i can be determined using experimental methods such as uniaxial, triaxial, and true triaxial compression tests, and direct and indirect tension tests. For the convenience of practical applications, the reference values of m_i for different rocks as tabulated in Table I were proposed in [5, 6]. Most of these values were derived on the basis of triaxial tests of intact rock samples.

Apart from the conceptual starting point provided by the Griffith theory, there is no fundamental relationship between the empirical parameter m_i in the H-B strength criterion and any physical characteristics of the intact rock. Zhang *et al.* [7] conducted numerical simulations with the 3D Particle Flow Code (PFC3D) to study the effect of microscopic parameters, particle size, particle size distribution, contact model, contact normal to shear strength ratio, and particle friction coefficient, on the strength parameter m_i by considering the intact rock as a cemented granular material consisting of spherical micro-particles. The results (Table II) indicate the following:

- The particle size has essentially no effect on m_i , and the influence of particle size can be simply ignored in the simulations.
- The different exponential size distributions realized by clumping balls into larger size particles to change the number of different size particles only slightly change the m_i values.
- The parallel-bond model that transmits both forces and moments between particles gives larger m_i values than the contact-bond model, which can only resist forces at the contact, when the same microscopic parameters are used. This is mainly due to the additional notional cement in the parallel-bond, which represents the granular interlock [8].
- The normal to shear strength ratio has modest effect on m_i when the shear strength is larger than the normal strength. Higher shear strength leads to larger intact rock strength and higher m_i value. However, m_i only changes slightly when the shear strength is smaller than the normal strength.
- The m_i increases slightly with higher particle friction coefficient.

Table I. Values of parameter m_i for different rocks [5, 6].

Rock type	Class	Group	Texture			
			Coarse	Medium	Fine	Very fine
Sedimentary	Clastic		Conglomerate (21 ± 3) ^a Breccia (19 ± 5)	Sandstone 17 ± 4	Siltstone 7 ± 2 Greywacke (18 ± 3)	Claystone 4 ± 2 Shale (6 ± 2) Marl (7 ± 2)
	Non-clastic	Carbonate Evaporite Organic	Crystalline limestone (12 ± 3)	Sparitic limestone (10 ± 2) Gypsum 8 ± 2	Micritic limestone (9 ± 2) Anhydrite 12 ± 2	Dolomite (9 ± 3)
Metamorphic	Non-foliated		Marble 9 ± 3	Hornfels (19 ± 4) Metasandstone (19 ± 3)	Quartzite 20 ± 3	Chalk 7 ± 2
	Slightly foliated Foliated ^b Plutonic	Light Dark	Migmatite (29 ± 3) Granite 32 ± 3 Granodiorite (29 ± 3) Gabbro 27 ± 3 Norite 20 ± 5 Porphyry (20 ± 5)	Schist 12 ± 3 Diorite 25 ± 5 Dolerite (16 ± 5)	Gneiss 28 ± 5 Phyllite (7 ± 3)	Slate 7 ± 4
Igneous	Hypabyssal Volcanic	Lava	Agglomerate (19 ± 3)	Rhyolite (25 ± 5) Andesite 25 ± 5 Breccia (19 ± 5)	Diabase (15 ± 5) Dacite (25 ± 3) Basalt (25 ± 5) Tuff (13 ± 5)	Peridotite (25 ± 5) Obsidian (19 ± 3)
		Pyroclastic				

^aValues in parenthesis are estimates.^bThese values are for intact rock specimen tests normal to bedding or foliation. The value of m_i will be significantly different if failure occurs along a weakness plane.

Table II. The results of m_i from numerical simulations with different microscopic parameters using spherical micro-particles (revised and extended from [7]).

Factor	Microscopic parameter				Obtained m_i			
	Case A	Case B	Case C	Case D	Case A	Case B	Case C	Case D
Particle size	1.02	Particle radius $R_{\min}$ 1.14	$R_{\max}/R_{\min} = 1.66$, unit: mm 1.26	1.38	2.45	2.46	2.46	2.44
Particle size distribution	Number of clumped large size particles 0/0/0	(radius = 5.4 mm/2.7 mm/1.35 mm) 2/4/8	4/8/16	8/16/32	2.45	2.38	2.39	2.60
Contact model	Contact bond Particle modulus 72	Parallel bond Particle modulus/parallel bond modulus (unit: GPa) 28/64	48/48	69/35	2.45	2.85	3.64	5.11
Contact normal and shear strengths	74/74	Contact normal strength/contact shear strength (unit: MPa) 56/112 122/61	56/168 183/61	56/224 244/61	2.45	5.06 2.08	8.26 2.25	10.13 2.39
Particle friction	0.4	Particle friction coefficient 0.5	0.6	0.7	2.11	2.45	2.96	3.52

The numerical specimens are set as a prism in shape, with a height 60 mm and a lateral length 30 mm.

In summary, although the contact model, the contact normal to shear strength ratio, and the particle friction coefficient can affect m_i in a certain level, the obtained m_i values from the numerical simulations using spherical micro-particles are always below 12, no matter what microscopic parameters were used. Thus, the numerical simulation with spherical micro-particles can only obtain m_i values within the lower range of reported m_i values between 2 and 35 (Table I). This is because high-level interlocking cannot be achieved between spherical micro-particles. The authors later tried different numerical simulations and found that higher m_i values can be obtained when non-spherical micro-particles are used. This is similar to what Cho *et al.* [9] did using clumped-particle geometry to predict the stress–strain behavior and the complete nonlinear failure envelope of Lac du Bonnet (LDB) granite with two-dimensional particle flow modeling (PFC2D). They pointed out that the default particle flow code cannot predict the failure envelope (which also means the m_i) measured on laboratory samples, unless the clumped-particle geometry is used.

In this paper, a systemic study of the effect of non-spherical micro-particles on the H–B strength parameter m_i is presented. Because the key for this study is to construct numerical specimens containing non-spherical micro-particles so that high-level interlocking between particles can be achieved, the methods for generating non-spherical micro-particles in PFC3D are first discussed in detail. Then, the numerical simulations are briefly described, and the results regarding the effect of non-spherical micro-particles on m_i are presented and discussed. Finally, two widely studied intact rocks, LDB granite and Carrara marble, are simulated to further demonstrate the importance to use appropriate non-spherical micro-particles to obtain the right m_i values.

2. CONSTRUCTION OF NUMERICAL SPECIMENS CONTAINING NON-SPHERICAL MICRO-PARTICLES IN PFC3D

To conduct numerical simulations to investigate the effect of non-spherical micro-particles on the strength parameter m_i , a key is to construct the numerical specimen containing the non-spherical micro-particles. Therefore, this section briefly describes the genesis process of numerical specimens containing non-spherical micro-particles in PFC3D.

There are two methods, direct and indirect, for genesis of non-spherical micro-particles. The direct method produces non-spherical particles directly, and the indirect method first produces spherical particles and then clumps the particles within specified volumes to form non-spherical particles. Thus far, the indirect method has been more widely used than the direct method because of two reasons. First, the model process and the detection of contact between non-spherical micro-particles are extremely complicated for the direct method, especially when the aspect ratio of the non-spherical micro-particles is greater than 2. Second, the basic particles of intact rock may include both spherical and non-spherical ones. The studies by Hornby *et al.* [10], Verhelst *et al.* [11], Wang *et al.* [12], and Chen *et al.* [13] with microscopic techniques indicate that a great portion of the large non-spherical micro-particles are in contact with each other and form the skeleton of intact rock. Some small micro-particles fill in the void space between the large non-spherical micro-particles. These small micro-particles can be regarded as spheres approximately. The indirect method can be applied to produce the required number of spherical and non-spherical micro-particles. Therefore, the indirect method is used in this study.

In PFC3D V4.0, the command ‘Clump’ is an indirect method for genesis of non-spherical micro-particles. This command makes the generation of non-spherical micro-particles quite convenient and efficient. A clump is a group of slaved balls that behave as a rigid body (with a deformable boundary) that will not break apart, regardless of the forces acting on it. A clump can serve as a super particle of general shape. PFC3D V4.0 provides two methods for genesis of clumps, Clump Replace and Clump Range. Clump Replace is a method that replaces the original particle with a stick clump of the same volume and centroidal location, and an arbitrary orientation. This method is simple and flexible in operation, but the orientation of the clump is uncontrollable and a larger volume may be generated. The Clump Range is a method that preinstalls the clump range with

specified shape, volume, centroidal location, and orientation. After a clump range is specified, the balls within that range are simply added in the clump. The Clump Range method is selected in this study.

2.1. Installation of clumps

For a specific simulation, the number, centroidal location, volume, shape, and orientation of clumps are first determined and added in a numerical model containing balls. The shape of basic micro-particles for real intact rock is irregular. Wang *et al.* [12] developed the equivalent ellipsoid method to represent the real particles and indicated that the irregular basic micro-particles can be reconstructed with ellipsoids at reasonable accuracy. Thus, in this paper, ellipsoids are used in the simulations to study the effect of non-spherical micro-particles on m_i .

To install clumps with random centroidal locations in a numerical specimen (Figure 1), it is important to ensure that the clumps do not intersect each other. This is achieved following an iteration process as shown in Figure 2. After the first clump is installed at a random centroidal location, the second clump is installed randomly. Then, a check is performed to check if the second clump intersects the first clump. If intersection happens, another second clump is installed randomly, and the intersection check is performed again. This process is continued until the second clump does not intersect the first one. Similarly, the third clump to be installed should not intersect the first and second clumps. The algorithms for determining the clump range and checking the clump intersection are discussed as follows.

(1) Basis for determination of clump range

To determine the spatial relationship between a newly installed clump range and the existing ones and check whether they intersect or not, the clump ranges need to be described with mathematical expressions. For a clump range in the shape of an ellipsoid, it can be described by the following mathematical expression:

$$\begin{aligned} & \frac{[(l_1)_i \cdot (x - x_i) + (m_1)_i \cdot (y - y_i) + (n_1)_i \cdot (z - z_i)]^2}{a_i^2} \\ & + \frac{[(l_2)_i \cdot (x - x_i) + (m_2)_i \cdot (y - y_i) + (n_2)_i \cdot (z - z_i)]^2}{b_i^2} \\ & + \frac{[(l_3)_i \cdot (x - x_i) + (m_3)_i \cdot (y - y_i) + (n_3)_i \cdot (z - z_i)]^2}{c_i^2} = 1 \end{aligned} \quad (5)$$

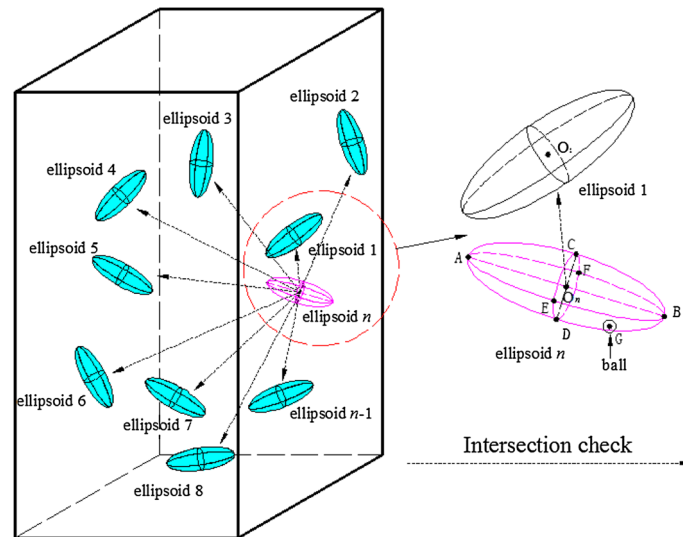


Figure 1. Genesis process of non-spherical (ellipsoid) particles.

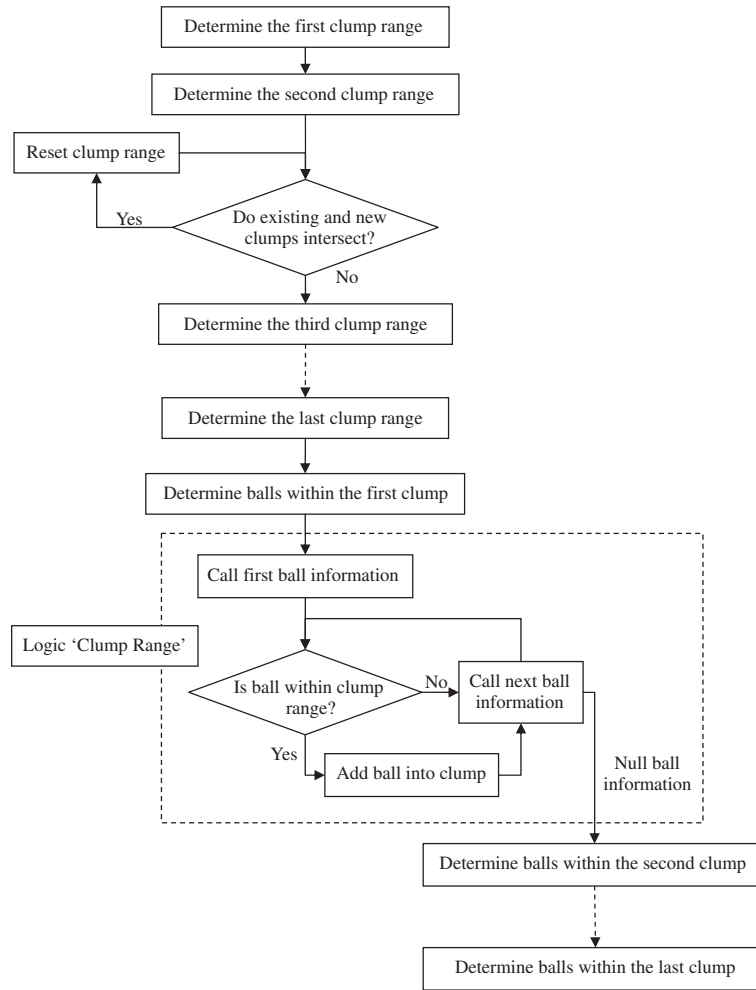


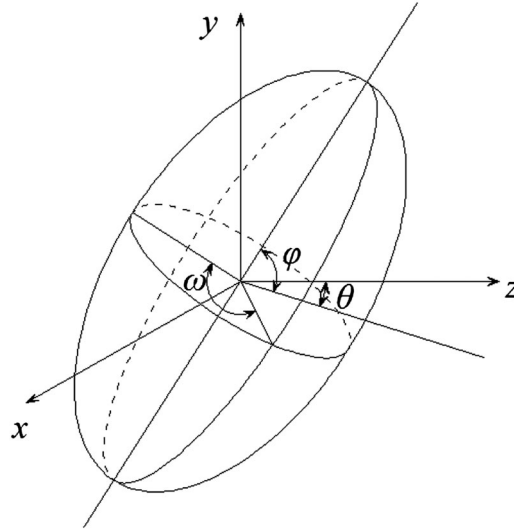
Figure 2. Flowchart for clump (non-spherical micro-particle) genesis process.

where x_i , y_i , and z_i are the coordinates of the centroidal location of ellipsoid i ($i = 1, 2, \dots, n$); a_i , b_i , and c_i are the lengths of the three semi-axes of the ellipsoid i ($b_i \geq c_i \geq a_i$); and $(l_1)_i$, $(l_2)_i$, $(l_3)_i$, $(m_1)_i$, $(m_2)_i$, $(m_3)_i$, $(n_1)_i$, $(n_2)_i$, $(n_3)_i$ are as follows:

$$\begin{cases} (l_1)_i = \sin\varphi_i \sin\omega_i \\ (l_2)_i = \sin\varphi_i \cos\omega_i \\ (l_3)_i = \cos\varphi_i \\ (m_1)_i = \cos\theta_i \cos\omega_i - \cos\varphi_i \sin\theta_i \sin\omega_i \\ (m_2)_i = -\cos\theta_i \sin\omega_i - \cos\varphi_i \sin\theta_i \cos\omega_i \\ (m_3)_i = \sin\varphi_i \sin\theta_i \\ (n_1)_i = \sin\theta_i \cos\omega_i + \cos\varphi_i \cos\theta_i \sin\omega_i \\ (n_2)_i^2 = -\sin\theta_i \sin\omega_i + \cos\varphi_i \cos\theta_i \cos\omega_i \\ (n_3)_i = -\sin\varphi_i \cos\theta_i \end{cases} \quad (6)$$

in which φ_i , θ_i , and ω_i are the three angles denoting the orientation of the ellipsoid i ($i = 1, 2, \dots, n$); φ_i and θ_i being the dip and dip direction angle of the main axial cross section; and ω_i the self-rotational degree (Figure 3).

Zhang *et al.* [7] proposed a method for determining the spherical clump ranges based on the distance between centroidal locations, but it is not applicable to ellipsoids. To determine if a newly installed

Figure 3. Three ellipsoid angles φ , θ , and ω .

ellipsoid intersects an existing one, an efficient method is proposed by considering only the six pole points of the newly installed ellipsoid. The three axial lines of a newly installed ellipsoid, say ellipsoid n , can be described by the following mathematical expressions:

$$\begin{aligned}\frac{x - x_n}{(l_1)_n} &= \frac{y - y_n}{(m_1)_n} = \frac{z - z_n}{(n_1)_n} \\ \frac{x - x_n}{(l_2)_n} &= \frac{y - y_n}{(m_2)_n} = \frac{z - z_n}{(n_2)_n} \\ \frac{x - x_n}{(l_3)_n} &= \frac{y - y_n}{(m_3)_n} = \frac{z - z_n}{(n_3)_n}\end{aligned}\quad (7)$$

The coordinates of the six pole points (A, B, C, D, E, and F) of the newly installed ellipsoid can be determined by solving Eqs. (5) and (7) as follows:

$$\begin{aligned}x_{A,B} &= \pm a_n(l_1)_n + x_n & x_{C,D} &= \pm b_n(l_2)_n + x_n & x_{E,F} &= \pm c_n(l_3)_n + x_n \\ y_{A,B} &= \pm a_n(m_1)_n + y_n & y_{C,D} &= \pm b_n(m_2)_n + y_n & y_{E,F} &= \pm c_n(m_3)_n + y_n \\ z_{A,B} &= \pm a_n(n_1)_n + z_n & z_{C,D} &= \pm b_n(n_2)_n + z_n & z_{E,F} &= \pm c_n(n_3)_n + z_n\end{aligned}\quad (8)$$

Then the coordinates of the six pole points (A, B, C, D, E, and F) can be substituted into the expression of the existing ellipsoids (1, 2, ..., $n-1$), for an example ellipsoid 1, to check if they intersect each other:

$$\begin{aligned}f(x_0, y_0, z_0) &= \frac{[(l_1)_1 \cdot (x_0 - x_1) + (m_1)_1 \cdot (y_0 - y_1) + (n_1)_1 \cdot (z_0 - z_1)]^2}{a_1^2} \\ &+ \frac{[(l_2)_1 \cdot (x_0 - x_1) + (m_2)_1 \cdot (y_0 - y_1) + (n_2)_1 \cdot (z_0 - z_1)]^2}{b_1^2} \\ &+ \frac{[(l_3)_1 \cdot (x_0 - x_1) + (m_3)_1 \cdot (y_0 - y_1) + (n_3)_1 \cdot (z_0 - z_1)]^2}{c_1^2}\end{aligned}\quad (9)$$

in which the subscript '0' represents A, B, C, D, E, and F, respectively. If the value of $f(x_0, y_0, z_0)$ in Eq. (9) is greater than 1, the newly installed ellipsoid does not intersect the existing ellipsoid. Meanwhile, to prevent the existing ellipsoids from intersecting the new one, a similar checking process is performed.

(2) Algorithm for determination of clump range

The determination of clump ranges consists of several sub-processes related to random installation and intersection check. To do that, an iteration following a loop algorithm needs to be carried out. There are several loop algorithms, such as randomized algorithm [14], divide and conquer algorithm [15, 16], and backtracking algorithm [17].

Although the randomized algorithm can lead to the maximum randomness of non-spherical micro-particle distributions, the iteration process does not terminate when the percentage of clumps is greater than 17% (which represents the number of balls within the non-spherical micro-particles relative to the total number of balls). The divide and conquer algorithm divides a numerical specimen into several small sub-ranges and then solves the problem in each sub-range. This algorithm needs too much manual intervention, and it is difficult to ensure the required randomness of non-spherical micro-particles distributions. The backtracking algorithm is essentially an improved randomized algorithm. By setting a threshold number for the iteration times to avoid endless iterations and allowing the loop to backtrack to the previous one to restart the process, the problem of endless iterations for the randomized algorithm is solved. On the basis of the efficiency of the process and the acceptability of the final results, the backtracking algorithm is used in this study.

Figure 4 shows the non-spherical particles (ellipsoids) on a cross section of numerical specimens containing respectively 10%, 20%, 30%, and 40% of clumps. It can be seen that the particles are uniformly distributed in space and do not intersect each other, showing the effectiveness of the method for installing clumps. Some of the non-spherical particles look like spheres on the cross section simply due to the relative direction between the particle and the cross section.

2.2. Determination of balls within a clump range

After a clump range is determined, the balls within the whole specimen are checked to determine if they are within the clump range, following the process shown in Figure 2. This can be carried out by checking the value of the related mathematical expressions. For example, to determine if ball G

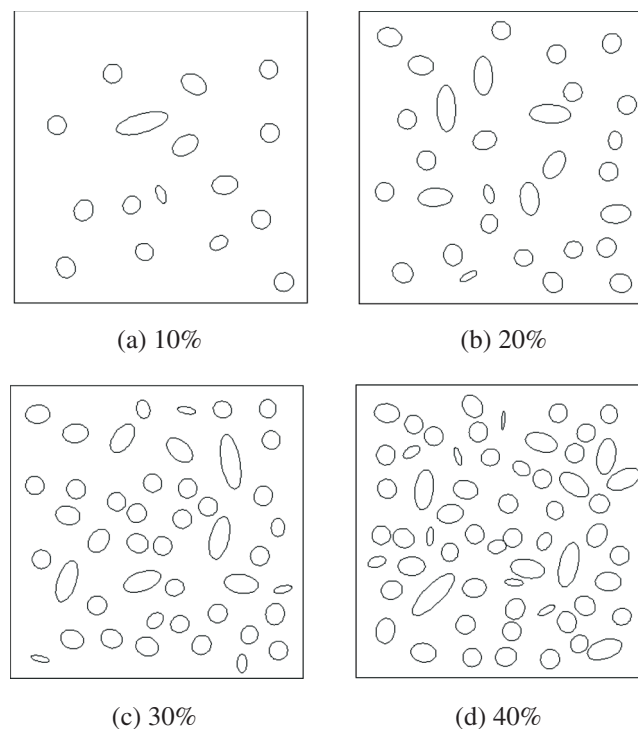


Figure 4. Cross sections showing non-spherical particles in numerical specimens containing different percentages of clumps.

is within clump (ellipsoid n) shown in Figure 1, Eq. (10) can be checked. If Eq. (10) is true, the ball is within the clump range.

$$\begin{aligned} & \frac{[(l_1)_n \cdot (x_G - x_n) + (m_1)_n \cdot (y_G - y_n) + (n_1)_n \cdot (z_G - z_n)]^2}{a_n^2} \\ & + \frac{[(l_2)_n \cdot (x_G - x_n) + (m_2)_n \cdot (y_G - y_n) + (n_2)_n \cdot (z_G - z_n)]^2}{b_n^2} \\ & + \frac{[(l_3)_n \cdot (x_G - x_n) + (m_3)_n \cdot (y_G - y_n) + (n_3)_n \cdot (z_G - z_n)]^2}{c_n^2} < 1 \end{aligned} \quad (10)$$

where a_n , b_n , and c_n are the lengths of the three semi-axes of clump range (ellipsoid n); x_G , y_G , z_G , and x_n , y_n , z_n are the centroidal location coordinates of the ball (point G) and the clump range (point O_n), respectively.

3. STUDY OF EFFECT OF NON-SPHERICAL MICRO-PARTICLES ON HOEK–BROWN STRENGTH PARAMETER M_I

For this study, the numerical specimens are set as a prism in shape, with a height 60 mm and a lateral length 30 mm. They are built following the standard PFC3D V4.0 modeling process, which has been described in detail by Itasca [18] and Zhang *et al.* [19] and included the steps as shown in Figure 5 (a)–(c). First, spherical micro-particles with radii satisfying a uniform particle-size distribution bounded by $R_{\min}=0.5$ mm and $R_{\max}=0.75$ mm are generated. Then, non-spherical micro-particles are generated following the procedure described in the previous section. Considering its simplicity,

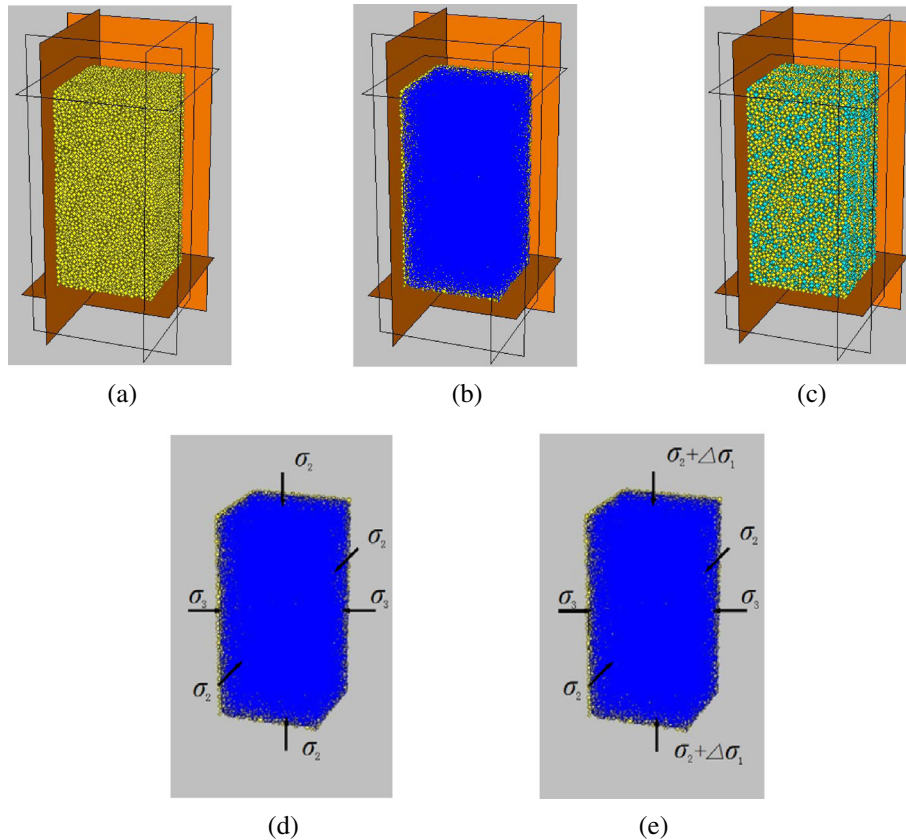


Figure 5. Numerical modeling and true triaxial testing process for intact rock specimen.

the contact–bond is used in this study. The same contact normal and shear strength of 150 MPa is used for all non-spherical micro-particles. The ball modulus and ball contact friction coefficient are set as 90 GPa and 0.5, respectively. The microscopic parameters coupled with deformation such as the normal to shear stiffness ratio have no obvious effect on the strength behavior and thus are set as the default values provided by the code. Because the main goal of this paper is to study the effect of non-spherical micro-particles on m_i , the micro mechanical properties are simply selected on the basis of the typical micro mechanical property values obtained from calibration of different types of rocks [19, 20].

Because triaxial compression tests have the limitation that the influence of the intermediate principal stress is not considered [2, 21, 22], true triaxial compression tests are simulated in this study. The numerical testing process is shown in Figure 5(d) and (e). For each condition, 10 numerical tests are conducted with different initial assembly of particles and non-spherical micro-particles, and the average results are used in the analysis. On the basis of the numerical true triaxial compression tests, the strength parameter m_i is determined by fitting with Eq. (2).

The study investigated the effect of number, aspect ratio, size, and shape of non-spherical micro-particles on m_i with systematically designed simulations (Table III). The results are presented and discussed as follows.

3.1. Effect of number of non-spherical micro-particles

The number of non-spherical micro-particles is represented by percentage (%), which is the number of balls within the non-spherical micro-particles relative to the total number of balls. In this study, 19%, 24%, 32%, 37%, and 42% are considered, and the 0% is added as a comparison. For all simulations, the aspect ratio is set as 1:3:1.

Figure 6 shows the obtained m_i at different number of non-spherical micro-particles. m_i increases with larger number of non-spherical micro-particles. When the number of non-spherical micro-particles is smaller than about 19%, the effect of non-spherical micro-particles is very small. This is mainly because when the number of non-spherical micro-particles is too small, say less than 19%, the non-spherical micro-particles have very few chances to be in touch with each other and provide interlocking between particles. When the number of non-spherical micro-particles is large enough, the non-spherical micro-particles will be in touch with each other and provide interlocking between particles. More non-spherical micro-particles lead to higher level of interlocking between particles and thus greater m_i . In next studies, for simplicity, the number of non-spherical micro-particles will be set as 32%.

3.2. Effect of aspect ratio of non-spherical micro-particle

Wang *et al.* [12] summarized the aspect ratios of 127 real micro-particles for two different types of intact rocks and indicated that most aspect ratios are within the range of 1:1:1 to 1:4:1. Thus, the aspect ratios of 1:1:1, 1:2:1, 1:2.5:1, 1:3:1, 1:3.5:1, and 1:4:1 are selected in the simulations to study the effect of aspect ratio, where 1 is a unit that equals 1.9 mm. The non-spherical micro-particles with different aspect ratios generated by the clump method are show in Figure 7. Figure 8 shows the simulation results. It can be seen that the strength parameter m_i is higher as the aspect

Table III. Selected values for studying the effect of non-spherical micro-particles on m_i .

		Values					Other three fixed parameters
Parameter		1	2	3	4	5	
A	Number	19%	24%	32%	37%	42%	B3, C3, D2
B	Aspect ratio	1:2:1	1:2.5:1	1:3:1	1:3.5:1	1:4:1	A3, C3, D2
C	Size	0.75:2.25:0.75	0.9:2.7:0.9	1:3:1	1.1:3.3:1.1	1.25:3.75:1.25	A3, B3, D2
D	Shape	1:3:0.75	1:3:1	1:3:1.5	1:3:2	1:3:2.5	A3, B3, C3

The percentage (%) represents the number of balls within the non-spherical micro-particles relative to the total number of balls (34 322). 1 unit is equal to 1.9 mm.

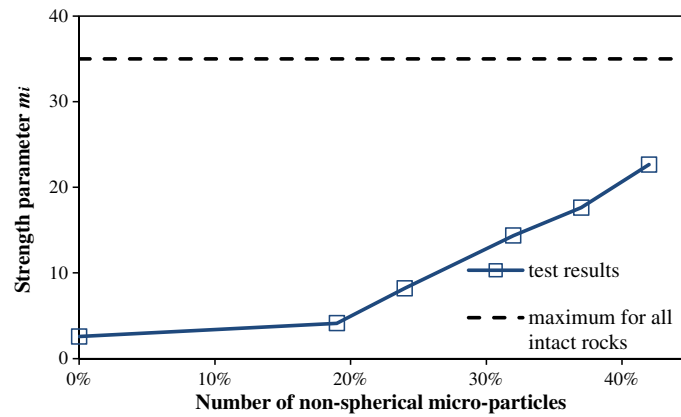


Figure 6. Numerical test results with different number of non-spherical micro-particles.

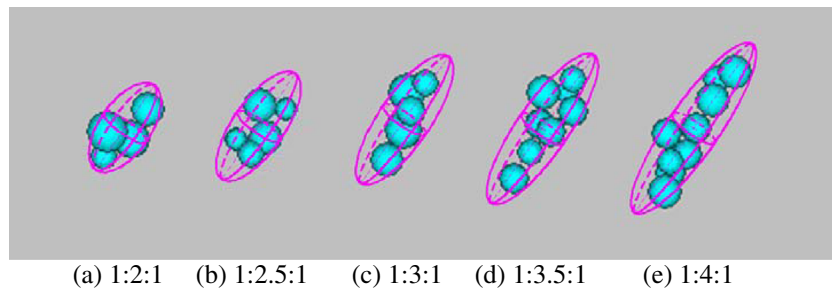


Figure 7. Non-spherical micro-particles with different aspect ratios generated by the clump method.

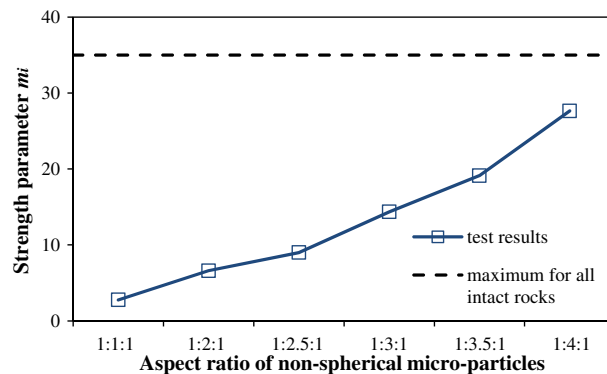


Figure 8. Numerical test results with different aspect ratios for the non-spherical micro-particles.

ratio of non-spherical micro-particles increases. This is mainly because larger aspect ratio leads to higher level of interlocking between particles.

3.3. Effect of size of non-spherical micro-particles

To study the effect of size, non-spherical micro-particles at 0.75:2.25:0.75, 0.9:2.7:0.9, 1:3:1, 1.1:3.3:1.1, and 1.25:3.75:1.25 are selected to conduct numerical simulations, where 1 is a unit that equals 1.9 mm. The non-spherical micro-particles with different sizes generated by the clump method are shown in Figure 9. The numerical test results are shown in Figure 10. It can be seen that the strength parameter m_i increases with larger size of non-spherical micro-particles. This is because

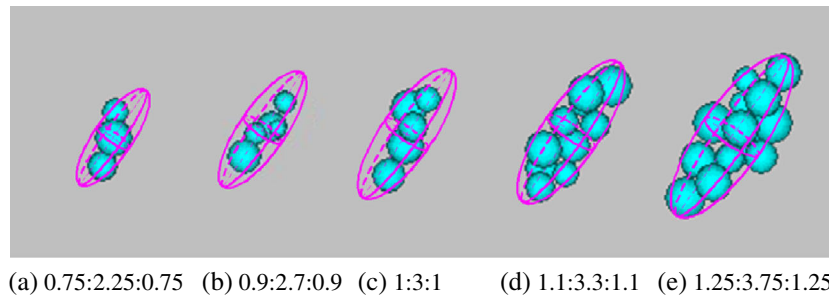


Figure 9. Non-spherical micro-particles with different sizes generated by the clump method.

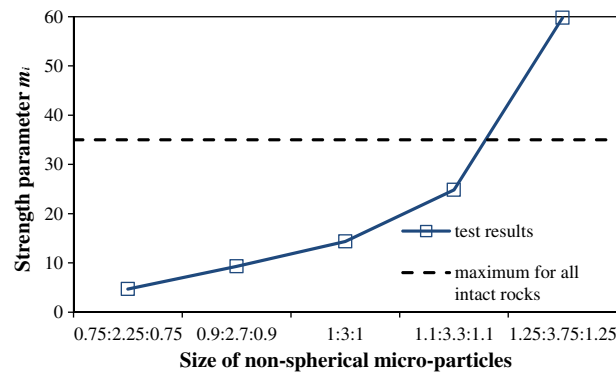


Figure 10. Numerical test results with different sizes of non-spherical micro-particles.

the movements between particles are likely to be blocked when the size of non-spherical micro-particles increases and the interlocking effect between particles is higher.

3.4. Effect of shape of non-spherical micro-particles

In the previous studies, the non-spherical micro-particles are ellipsoid with two of the three semi-axes having the same length. For some rocks, the particles are more like an ellipsoid with three semi-axes having different lengths. Thus, particles at 1:3:0.75, 1:3:1, 1:3:1.5, 1:3:2, and 1:3:2.5 are selected to conduct numerical simulations to study the effect of shape of non-spherical micro-particles, where 1 is a unit equal to 1.9 mm. Figure 11 shows non-spherical micro-particles of different shapes generated by the clump method. Figure 12 shows the simulation results. It can be seen that the strength parameter m_i increases when the non-spherical micro-particles become flatter. This is because flatter micro-particles lead to higher level of interlocking between particles. It is noted that micro-particles with an ellipsoid shape are simplified and commonly used models for representing the particles of rock. The true particle shape of rock may be significantly different. Other types of irregular shapes may be considered in further study.

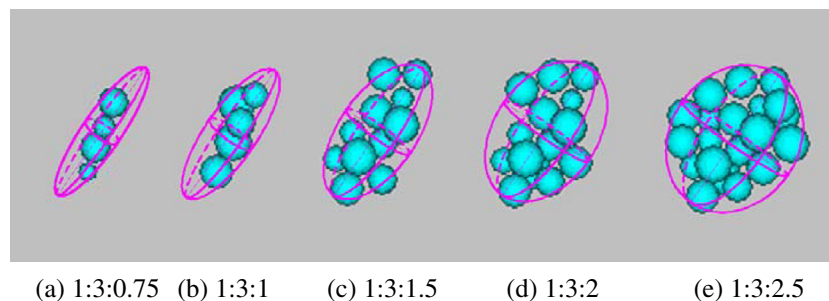


Figure 11. Non-spherical micro-particles with different shapes generated by the clump method.

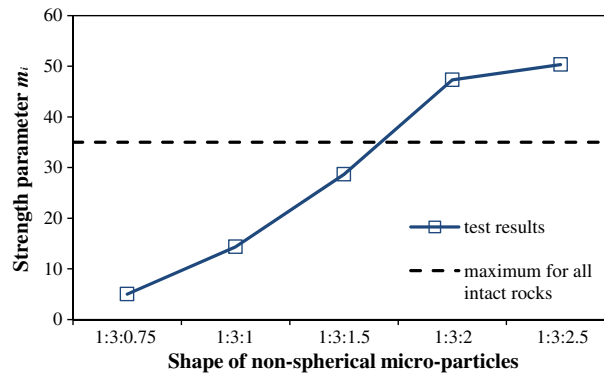


Figure 12. Numerical test results with different shapes of non-spherical micro-particles.

4. NUMERICAL SIMULATIONS OF TWO INTACT ROCKS

In this section, the particle flow modeling is used to simulate two widely studied intact rocks, LDB granite and Carrara marble. The results further show the importance of selecting appropriate micro-particles when using the particle flow modeling to simulate the behavior of intact rocks.

4.1. Numerical simulation of LDB granite

The intact LDB granite has a density of 2630 kg/m^3 , a uniaxial compressive strength of 200 MPa, and a Young's modulus of 69 GPa [8]. The particles of LDB granite have been studied by several researchers. Martin [23] stated that the grain size of LDB granite taken from the 240 Level of the URL (Underground Research Laboratory) ranges from 3 to 7 mm, with an average of 5 mm. Martin *et al.* [24] reported the grain size of LDB granite from the 420 Level of the URL ranging from 1 to 3.5 mm. As for the aspect ratio of the grains of LDB granite, no direct study can be found from the published literature, but some studies on the aspect ratio of granite grains can be used for a reference. Schulmann *et al.* [25] stated that the aspect ratio is 2.0 for quartz grain, which consists of 35% of granite. Grégoire *et al.* [26] stated that the average aspect ratio for the basic grain of granite is about 2.0, with the maximum being greater than 4.0. Nishiyama *et al.* [27] used a grain aspect ratio from 2.0 to 5.0 to study the microscopic behavior of Inada granite subject to a triaxial compression test. Thus, to simulate the LDB granite, an aspect ratio of 3.0 was selected for the non-spherical micro-particles. The non-spherical micro-particles were generated following the procedure presented earlier by first generating spherical micro-particles satisfying a uniform particle-size distribution bounded by $R_{\min} = 0.5 \text{ mm}$ and $R_{\max} = 0.75 \text{ mm}$. The non-spherical micro-particles have an average size of 3.5 mm. Considering the spherical and non-spherical micro-particles together, the average particle size is in the range of 1.9–2.1 mm, which varies with the number of non-spherical

Table IV. Microscopic parameters and numerical test results of specimens A, B, and C for LDB granite and specimens D, E, and F for Carrara marble.

Specimen	Number of non-spherical micro-particles	Ball modulus (GPa)	C_B normal strength (MPa)	C_B shear strength (MPa)	Particle friction	Fitting m_i	Coefficient of determination (R^2)
A	30.7%	105	161	161	0.5	29.1	0.9347
B	32.3%	107	157	157	0.5	31.9	0.9308
C	33.9%	110	154	154	0.5	34.8	0.9345
D	—	27	52	135	0.4	5.97	0.9853
E	—	28	50	179	0.5	8.83	0.9952
F	—	28	49	285	0.7	11.8	0.9901

The number of non-spherical micro-particles is expressed as the percentage (%) of the balls within the non-spherical micro-particles relative to the total number of balls. The coefficient of determination (R^2) indicates how well the simulated true triaxial compressive test data is fitted using Eq. (2), 1 indicating a perfect fitting.

micro-particles. The numerical specimens are set as a prism in shape, with a height 60 mm and a lateral length 30 mm and containing 34 322 balls.

From Table I, the m_i for granite is in a range of (32 ± 3) . Thus, three different numerical specimens, A, B and C, containing different number of non-spherical micro-particles, were selected to simulate the

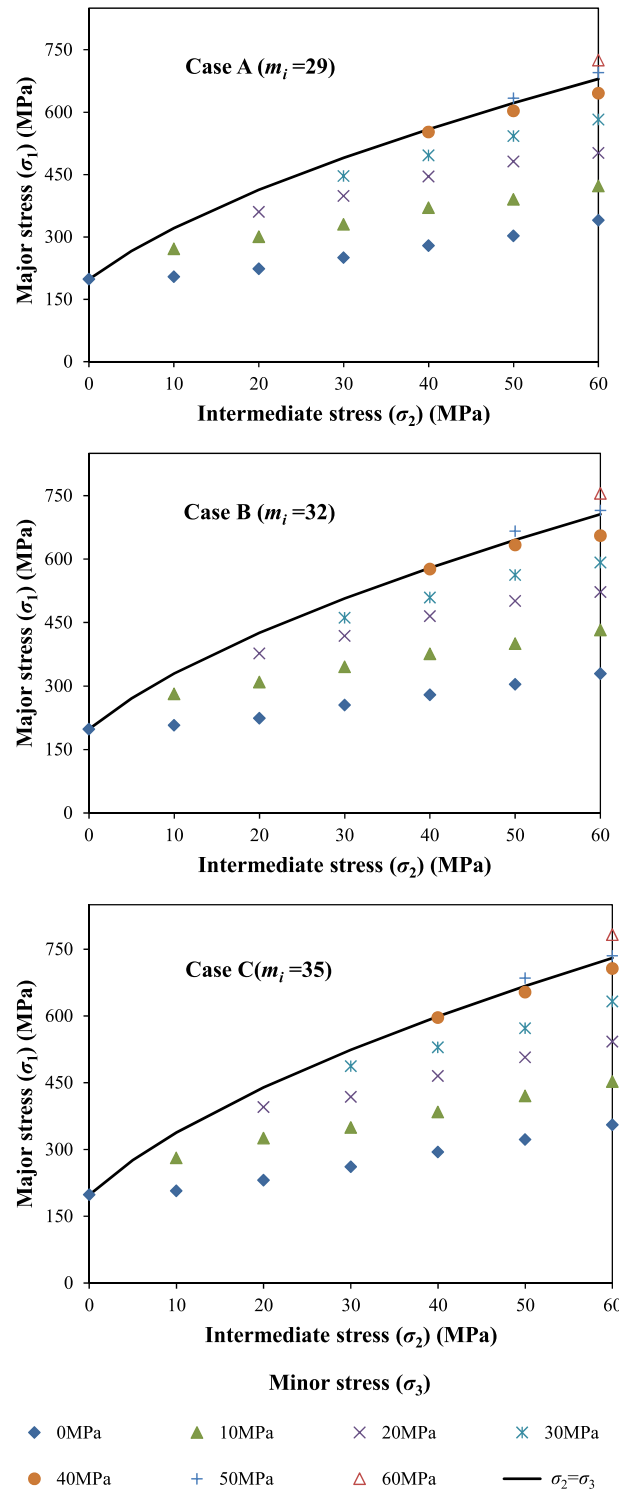


Figure 13. Numerical true triaxial test results of cases A, B, and C specimens for LDB granite. The solid lines are the H-B strength criterion lines ($\sigma_2 = \sigma_3$) at different values of m_i used in the simulation.

intact granite and give respectively the lower, average, and upper bound values of m_i for granite (Table IV). The three numerical specimens, A, B and C, contain 597, 632, and 663 non-spherical micro-particles, respectively. The microscopic parameters of these specimens were determined on

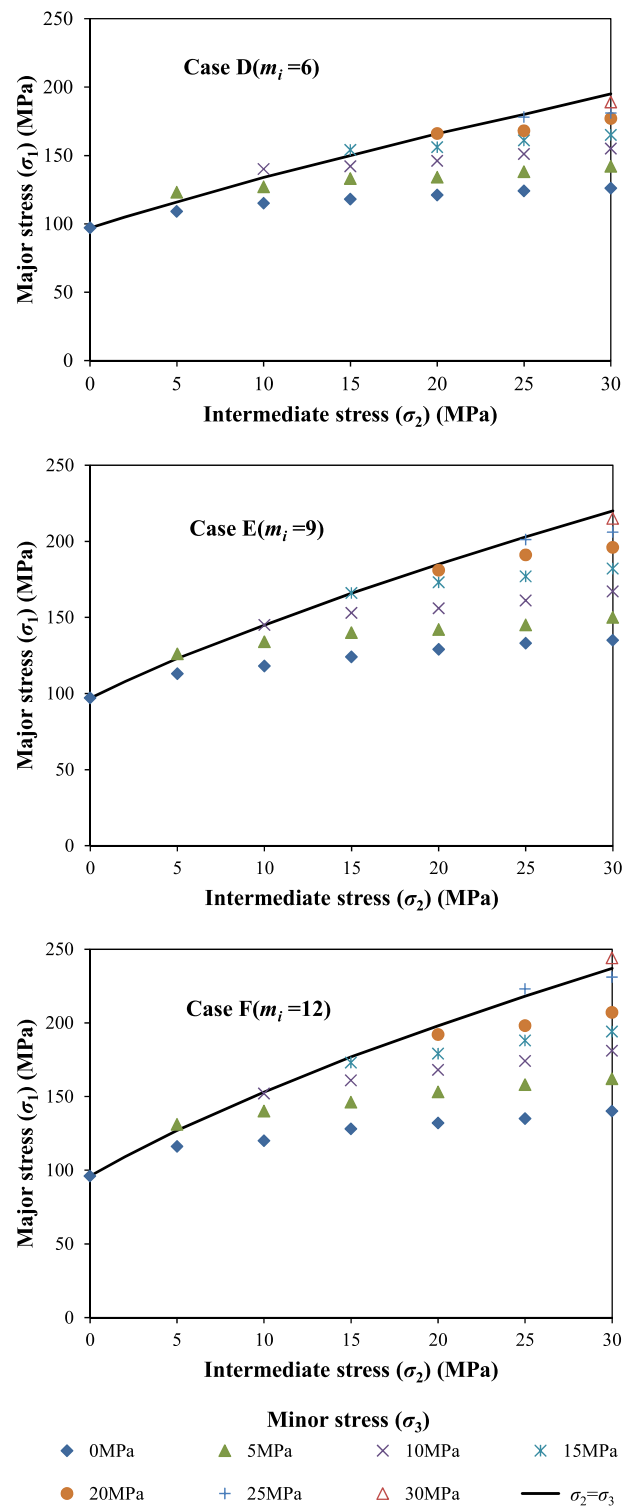


Figure 14. Numerical true triaxial test results of cases D, E, and F specimens for Carrara marble. The solid lines are the H-B strength criterion lines ($\sigma_2 = \sigma_3$) at different values of m_i used in the simulation.

the basis of simulations of uniaxial compression test to reproduce the strength and elastic modulus values of the LDB granite.

Figure 13 shows the results of numerical true triaxial compression tests on these three specimens. For a more direct comparison, Figure 15(a) shows the numerical test results and the predictions from the 3D H-B strength criterion (Eq. (2)) of Zhang and Zhu [2] with the simulated m_i values in

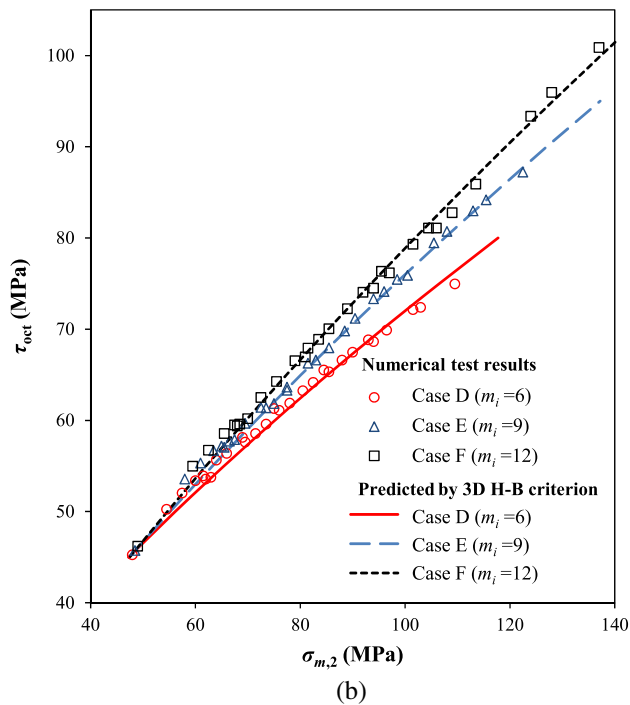
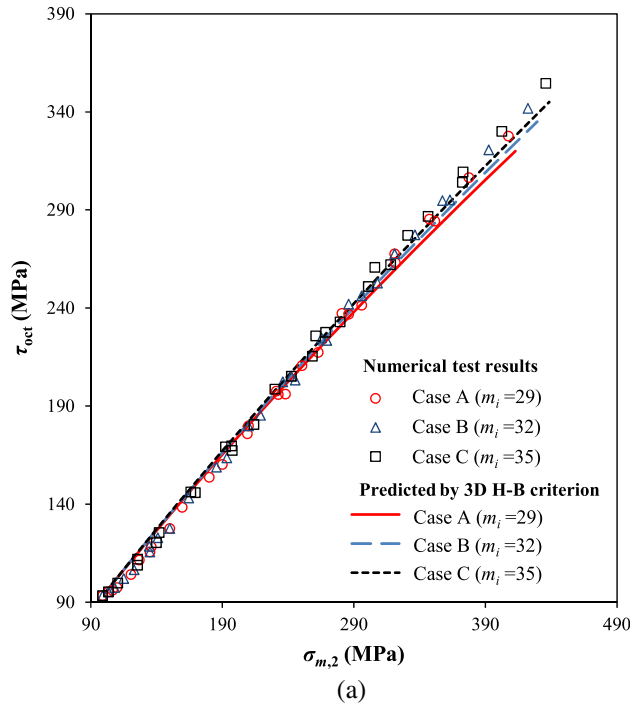


Figure 15. Compression of numerical true triaxial test results with 3D H-B strength criterion predictions: (a) LDB granite, and (b) Carrara marble.

$\sigma_{m,2}$ versus τ_{oct} . It can be seen that the numerical true triaxial test results are in good agreement with the 3D H-B strength criterion predictions.

4.2. Numerical simulation of Carrara marble

The intact Carrara marble has a density of 2560 kg/m³, a uniaxial compressive strength of 99.5 MPa, and a Young's modulus of 17.2 GPa [28]. The particles of Carrara marble and other types of marble have been studied by several researchers. The average aspect ratio of individual grains is 1.5 for the Carrara marble from two locations in Italy [29]. De Bresser *et al.* [30] used scanning electron microscopy to study the microstructure and obtained the aspect ratios of marble grains under different temperature and moisture conditions, which are in the range from 1.04 to 1.57. From the previous section, the effect of aspect ratio of non-spherical micro-particles is small when it is less than 2.0. Thus, for the numerical simulation of Carrara marble, spherical micro-particles were selected without considering the effect of non-spherical micro-particles.

The research data indicate that the average grain diameter is 1.25 mm for Mito marble [31], 3.5 mm for Yamaguchi marble [32], and 1.9 mm for Georgia marble [33]. Thus, in the numerical simulations, spherical micro-particles with diameters satisfying a uniform particle-size distribution bounded by $R_{\min} = 0.7$ mm and $R_{\max} = 1.05$ mm were selected. The numerical specimens for Carrara marble were set as a prism in shape, with a height 60 mm and a lateral length 30 mm and containing 12 508 balls.

From Table I, the m_i for marble is in a range of (9 ± 3) . Thus, three numerical specimens, D, E, and F, containing spherical micro-particles were selected to simulate the intact marble and give respectively the lower, average, and upper bound values of m_i for marble (Table IV). Again, the microscopic parameters of these specimens were determined on the basis of simulations of uniaxial compression test to reproduce the strength and elastic modulus values of Carrara marble. Figure 14 shows the numerical true triaxial test results at the three m_i values, and Figure 15(b) compares the numerical test results and the predictions using the 3D H-B strength criterion of Zhang and Zhu [2] with the simulated m_i values in $\sigma_{m,2}$ versus τ_{oct} . Again, the simulation results and the 3D H-B strength criterion predictions are in good agreement.

5. CONCLUSIONS

The effect of non-spherical micro-particles on the H-B strength parameter m_i was studied on the basis of numerical true triaxial tests using PFC3D V4.0. The following conclusions can be drawn on the basis of the obtained results:

1. The number, aspect ratio, size and shape of non-spherical micro-particles have a great effect on the simulated H-B strength parameter m_i . The strength parameter m_i increases when the number, aspect ratio, or size is larger or the shape becomes more irregular, mainly as a result of the higher level of interlocking between particles. This confirms the observations from engineering experience and laboratory experiments.
2. To simulate the right strength parameter m_i , numerical specimens containing appropriate non-spherical micro-particles need to be used. The genesis process and the loop algorithm proposed in this paper can be used to produce the specimens with sufficient efficiency.
3. The simulation results of two widely studied intact rocks, LDB Granite and Carrara Marble, are in good agreement with the 3D H-B strength criterion predictions and further indicate the importance of non-spherical micro-particles in controlling the behavior of intact rock.

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REFERENCES

1. Hoek E, Brown ET. Empirical strength criterion for rock masses. *Journal of Geotechnical and Geoenvironmental Engineering* 1980; **106**(9):1013–1035.
2. Zhang L, Zhu H. Three-dimensional Hoek–Brown strength criterion for rocks. *Journal of Geotechnical and Geoenvironmental Engineering* 2007; **133**(9):1128–1135.
3. Mogi K. Fracture and flow of rocks under high triaxial compression. *Journal of Geophysical Research* 1971; **76**(5): 1255–1269.
4. Hoek E, Brown ET. *Underground Excavations in Rock*. Institute Mining and Metallurgy: London, 1980.
5. Marinos P, Hoek E. Estimating the geotechnical properties of heterogeneous rock masses such as flysch. *Bulletin of Engineering Geology and the Environment* 2001; **60**(2):85–92.
6. Hoek E, Brown ET. Practical estimates of rock mass strength. *International Journal of Rock Mechanics and Mining Sciences* 1997; **34**(8):1165–1186.
7. Zhang Q, Zhu H, Zhang L, Ding X. Effect of micro-parameters on the Hoek–Brown strength parameter m_i for intact rock using particle flow modeling. *46th US Rock Mechanics Geomechanics Symposium*, Chicago, v3, 2012; 2187–2193.
8. Potyondy DO, Cundall PA. A bonded-particle model for rock. *International Journal of Rock Mechanics and Mining Sciences* 2004; **41**(8):1329–1364.
9. Cho N, Martin CD, Sego DC. A clumped particle model for rock. *International Journal of Rock Mechanics and Mining Sciences* 2007; **44**(7):997–1010.
10. Hornby BE, Schwartz LM, Hudson JA. Anisotropic effective-medium modeling of the elastic properties of shales. *Geophysics* 1994; **59**(10):1570–1583.
11. Verhelst F, Vervoort ADB, Marchal G. X-ray computerized tomography: determination of heterogeneities in rock samples. *8th ISRM Congress*, A. A. Balkema, Tokyo, Japan, 1995; 105–108.
12. Wang LB, Park JY, Fu Y. Representation of real particles for DEM simulation using X-ray tomography. *Construction and Building Materials* 2007; **21**(2):338–346.
13. Chen Y, Nishiyama T, Ito T. Application of image analysis to observe microstructure in sandstone and granite. *Resource Geology* 2001; **51**(3):249–258.
14. Rabin MO. Probabilistic algorithm for testing primality. *Journal of Number Theory* 1980; **12**(1):128–138.
15. Brassard G, Bratley P. *Fundamentals of algorithms*. Prentice Hall: New Jersey, 1996.
16. Levitin A. *Introduction to the Design and Analysis of Algorithms*. Addison-Wesley: New York, 2003.
17. Cormen TH, Leiserson CE, Rivest RL. *Introduction to Algorithms*. MIT Press: Cambridge, 1990.
18. Itasca. *PFC3D Particle Flow Code in 3-Dimensions, Version 4.0, User's Manual*. Itasca Consulting Group, Inc: Minneapolis, MN, 2008.
19. Zhang Q, Zhu H, Zhang L, Ding X. Study of scale effect on intact rock strength using particle flow modeling. *International Journal of Rock Mechanics and Mining Sciences* 2011; **48**(8):1320–1328.
20. Ding X, Zhang L, Zhu H, Zhang Q. Effect of model scale and particle size distribution on PFC3D simulation results. *Rock Mechanics and Rock Engineering* 2013. doi:10.1007/s00603-013-0533-1.
21. Zhang L. A generalized three-dimensional Hoek–Brown strength criterion. *Rock Mechanics and Rock Engineering* 2008; **41**(6):893–915.
22. Zhang Q, Zhu H, Zhang L. Modification of a generalized three-dimensional Hoek–Brown strength criterion. *International Journal of Rock Mechanics and Mining Science* 2013; **59**:80–96.
23. Martin CD. The strength of massive Lac du Bonnet granite around underground openings. *PhD thesis*, University of Manitoba, 1993.
24. Martin CD, Read RS, Martino JB. Observations of brittle failure around a circular test tunnel. *International Journal of Rock Mechanics and Mining Sciences* 1997; **34**(7):1065–1073.
25. Schulmann K, Mlčoch B, Melka R. High-temperature microstructures and rheology of deformed granite, Erzgebirge, Bohemian Massif. *Journal of Structural Geology* 1996; **18**(6):719–733.
26. Grégoire V, Darrozes J, Gaillot P, Nédélec A, Launeau P. Magnetite grain shape fabric and distribution anisotropy vs rock magnetic fabric: a three-dimensional case study. *Journal of Structural Geology* 1998; **20**(7):937–944.
27. Nishiyama T, Chen Y, Kusuda H, Ito T, Kaneko K, Kita H, Sato T. The examination of fracturing process subjected to triaxial compression test in Inada granite. *Engineering Geology* 2002; **66**(3–4):257–269.
28. Kovári K, Tisa A. Multiple failure state and strain controlled triaxial tests. *Rock Mechanics* 1975; **7**(1):17–33.
29. Molli G, Heilbronner R. Microstructures associated with static and dynamic recrystallization of Carrara marble (Alpi Apuane, NW Tuscany, Italy). *Geologie en Mijnbouw* 1999; **78**(1):119–126.
30. De Bresser JHP, Urai JL, Olgaard DL. Effect of water on the strength and microstructure of Carrara marble axially compressed at high temperature. *Journal of Structural Geology* 2005; **27**(2):265–281.
31. Mogi K. Deformation and fracture of rocks under confining pressure (2) elasticity and plasticity of some rocks. *Bulletin of the Earthquake Research Institute* 1965; **43**(2):349–379.
32. Mogi K. Deformation and fracture of rocks under confining pressure (1) compression tests on dry rock sample. *Bulletin of the Earthquake Research Institute* 1964; **42**(3):491–514.
33. Olsson WA. Grain size dependence of yield stress in marble. *Journal of Geophysical Research* 1974; **79**(32): 4859–4862.